

Tushar Bhattacharya

Gen AI Software Engineer — [linkedin.com/in/tushar-bhattacharya-68a210150](https://www.linkedin.com/in/tushar-bhattacharya-68a210150)
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Profile

Python Developer with more than 4 years of experience in designing, implementing, and maintaining scalable infrastructure solutions. Proficient in leveraging automation tools, cloud technologies, and best practices to streamline development, deployment, and operations processes. Adept at collaborating with cross-functional teams to drive continuous improvement and deliver high-quality, reliable software solutions.

Education

Bachelor of Technology — BTech, Information Technology

Jabalpur Engineering College, Jabal

Jan 2017 – Jan 2021

Skills

- Python (OOP, Flask, FastAPI, Flask), Git Version Control REST API development
- MySQL, Iceberg, MongoDB(NoSQL) User authentication
- ReactJS , User Authentication, Pyspark Async processing
- Full-stack integration - AI-ready data services AWS Cloud automation: AWS Lambda & CDK
- Python (boto3): API development Serverless architecture
- Infrastructure as Code (IaC), Logging, Observability, High-scale optimization Tool use simulation: JSON
- LLM prompt/data engineering, Function calling, Multi-turn dialogue generation API logic modeling, Instruction tuning

Employment

Software Engineer Wissen Technology

Sep 2024 – Present

1. Enabled seamless integration of structured and semi-structured datasets into downstream GenAI training workflows by handling transformation, normalization, and storage of event-triggered data batches.
2. Focused on preparing clean, audit-ready datasets for use in instruction tuning, model evaluation, or retrieval-augmented generation (RAG) pipelines.
3. Developed a data ingestion pipeline using AWS SNS, S3, and PySpark to automate event-triggered transformations and stored processed data in Iceberg tables.
4. Implemented logging of S3 event metadata for auditing, ensured system reliability with robust error handling along with handling file data transformations.

5. Cloud Development and automation using Python(boto3), AWS Lambda, Amazon S3, DynamoDB, AWS CDK(Cloud Development Kit), Load Balancer, Kubernetes and AWS SAM(Serverless Application Model) to create the infrastructure of the service. Also created lambdas to automate business critical task using AWS services.
6. Enabled end to end monitoring of the ingestion flow and transformations for the data using AWS Cloudwatch and Graphana dashboards for each file ingestion.

Generative AI Developer (Python LLM Trainer) Turing (Hourly Basis)

Apr 2024 – Present

1. Developed high-quality proprietary training datasets for foundational LLM companies, creating both SFT (Supervised Fine-Tuning) and RLHF (Reinforcement Learning from Human Feedback) datasets to enhance model performance and evaluation capabilities.
2. Authored multi-turn conversational datasets simulating realistic software engineering scenarios and user-AI assistant interactions, incorporating Python code generation, debugging workflows, and technical problem-solving patterns.
3. Conducted comprehensive model evaluations (Evals) by comparing LLM-generated outputs for correctness, code quality, and adherence to software engineering best practices, providing structured feedback for model alignment.
4. Developed & deployed serverless apps with AWS CDK (Cloud Development Kit) to define and provision cloud infrastructure, enhancing deployment speed and maintainability, AWS Lambda with API Gateway & RESTful APIs to automate process, improved efficiency by 98%.
5. Collaborated with cross-functional teams including prompt engineers and researchers on RLHF-based data generation tasks, ensuring datasets met benchmark-grade quality standards for model training and evaluation.
6. Created a full stack tool Cloudwatch log Fetcher tool to fetch the logs and based on use-cases filtering it so that the support engineers can use it to provide manual support to the advertiser for issues related to ironmountain website.
7. Designing(high level architecture), Implementing new features based on requirements, creating the workflow, utilizing AWS services and internal tools and ReactJS for the Ironmountain website(Legacy Service) by World Wide advertising Services(Amazon) for the advertiser to host their advertisement.

Python Developer (DevOps Engineer) Amazon

Sep 2022 – Apr 2024

1. Designed and implemented scalable features for the information legacy platform (advertising-facing site) using ReactJS, AWS services, and internal tooling, including high-level architecture, workflow design, and system enhancements.
2. Built time-range-based payment export functionality to Excel, improved codebase reliability via bug fixes, security patches, and performance optimizations for large-scale advertiser workflows.
3. Engineered serverless infrastructure using AWS CDK, Lambda, S3, DynamoDB, Load Balancer, and Kubernetes, enabling scalable cloud-native deployments and supporting business-critical pipelines.
4. Automated essential advertiser operations using Python (boto3), AWS SAM, and Lambda, improving system efficiency by 98% through infrastructure-as-code and self-healing serverless patterns.
5. Created and deployed a full-stack CloudWatch Log Fetcher tool that empowered support engineers to rapidly resolve real-time advertiser issues by filtering logs based on structured queries.
6. Contributed to reliability and maintainability of AI-driven advertising systems through infrastructure automation, operational tooling, and observability solutions supporting real-time advertiser engagement.

System Engineer Tata Consultancy Services

Jul 2021 – Sep 2022

1. Leveraged OOP principles in Python to develop backend logic and scalable services, while using ReactJS to deliver intuitive user-facing features, leading to a 90% improvement in user satisfaction on internal platforms.
2. Integrated PDF reader LLM with the website which helps the users with their use case specific queries fine tuned for each users using RAG.

3. Designed and optimized MySQL stored procedures for implementing **Slowly Changing Dimension (SCD) Type 2**, ensuring accurate historical tracking for master data used in downstream ML/AI pipelines.
4. Collaborated with product leadership to translate business models into production-grade web features, integrating frontend (ReactJS) with Python-based APIs (Flask, FastAPI).
5. Optimized the codebase with bug fixes, performance improvements, and security enhancements while implementing features like a time-range-based payment details export to Excel for users.
6. Built and maintained RESTful APIs with robust **user authentication, data validation, and asynchronous workflows**, enabling modular integration with data annotation, analytics, or LLM-support tools.
7. Ensured that back-end services were scalable, testable, and suitable for integration into **LLM evaluation platforms, fine-tuning pipelines, or RAG-based applications** that require dynamic data extraction and audit tracking from user information documents.

Projects

Database-Query-Generator-And-Data-Fetcher

Aug 2024 – Aug 2024

An AI-powered application that allows users to interact with events, company, and people data; identify companies attending events, find people working for companies attending events, and other data-related queries in layman language.

Retrieval-Augmented Generation (RAG) chatbot

May 2025 – May 2025

RAG-based chatbot for customer queries using predefined documents (medical insurance, policy rules, etc.).

Achievements

1. **Volunteered as an educator for Anganwadi (Center No. 18)** students in Jabalpur, focusing on foundational learning, engagement, and educational support for underprivileged children.
2. **Served as Coordinator for a Physics Workshop** in Jabalpur Engineering College, overseeing event planning, participant engagement, and logistical coordination for academic sessions and hands-on activities.

Certificates

- Front-End Web UI Frameworks and Tools: Bootstrap 4 (with Honors) Oct 2019
- Python 3 Programming Specialization Jun 2020
- Python Data Products for Predictive Analytics Specialization Jun 2020
- Python for Everybody Specialization Jun 2020
- Developing On AWS (Amazon) Feb 2024
- Learn key Data Engineering Skills such as SQL, Python, Apache Spark (Spark SQL and Pyspark) with Exercises and Projects Dec 2024
- AWS Developer Associate Certification Jan 2025
- Generative AI Foundations Certificate Aug 2025

Languages

Hindi, English

Hobbies

Football, Sketching